Sample Knowledge Document

PDF for Testing a Retrieval-Augmented Generation (RAG) System

1. What Is Machine Learning?

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every situation. A machine-learning system learns from examples and uses the learned patterns to process new data.

2. What Is Supervised Learning?

Supervised learning is a type of machine learning in which a model learns from labelled examples. Each training example contains input data and a known target output. The model learns a relationship between inputs and outputs, then uses that relationship to predict outputs for new examples. Classification and regression are common supervised-learning tasks.

3. What Is Unsupervised Learning?

Unsupervised learning works with data that does not have labelled target outputs. The algorithm attempts to discover patterns or structure in the data. Clustering is a common unsupervised-learning task; for example, a clustering algorithm may group customers with similar purchasing behaviour.

4. What Is Classification?

Classification is a machine-learning task that predicts a category or class. Examples include classifying an email as spam or not spam and predicting whether a customer will leave a service. Logistic regression, decision trees, and support vector machines can be used for classification.

5. What Is Regression?

Regression is a machine-learning task that predicts a numerical value. Examples include predicting house prices, product demand, or monthly sales. Linear regression and some tree-based models can be used for regression.

6. What Is Overfitting?

Overfitting occurs when a model learns the training data too closely, including noise or accidental patterns. It may perform very well on training data but poorly on unseen data. Overfitting can be reduced with techniques such as cross-validation, regularization, simpler models, and more representative training data.

7. What Is a Train-Test Split?

A train-test split divides a dataset into separate subsets. The training set is used to fit the model, while the test set is used to estimate how well the model performs on data it has not seen during training. The test set should not be used to make repeated tuning decisions.

8. What Is Deep Learning?

Deep learning is a subset of machine learning that uses neural networks with multiple layers to learn complex patterns. It is commonly used in image recognition, speech processing, language tasks, and other applications involving large or complex datasets.

9. What Is Natural Language Processing?

Natural language processing (NLP) is a field of artificial intelligence focused on enabling computers to process and work with human language. Common NLP tasks include text classification, sentiment analysis, information extraction, translation, and question answering.

10. What Is a Vector Embedding?

A vector embedding is a numerical representation of data, such as text, that is designed to capture useful semantic or contextual relationships. In a retrieval system, embeddings can help identify text passages that are similar in meaning to a user's question, even when the wording differs.

11. What Is RAG?

Retrieval-Augmented Generation (RAG) is an approach that retrieves relevant information from an external source and provides it to a language model as context for generating an answer. A typical document RAG pipeline extracts text, divides it into chunks, creates embeddings, retrieves relevant chunks for a question, and passes those chunks to a language model.

12. What Is a Vector Database or Index?

A vector index stores or organizes vector representations so that similar vectors can be searched efficiently. FAISS is a library for similarity search over dense vectors. It can be used to retrieve document chunks whose embeddings are similar to a question embedding.

Questions to Test a PDF RAG System

- What is supervised learning?
- How is classification different from regression?
- What is overfitting and how can it be reduced?
- What is RAG?
- What does a vector embedding represent?
- What is the capital of Japan? (This information is not included in this document.)